



How can the next solar cycle be predicted?

Data Analysis of Solar Cycles

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Introduction

The solar cycle is an 11-year recurring event that flips the Sun's North and South poles. Scientists keep track of the repeating cycles by tracking the sunspots as it may intervene with the workings of Earth. In this project, available data set was downloaded, average length and peak volume was measured and three future cycles were predicted. The purpose of this project is to present given data from a dataset and make conclusions based on the research question.

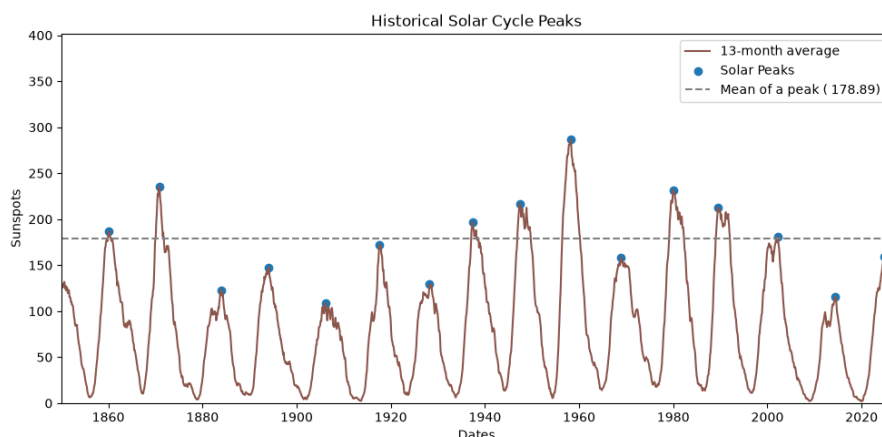
Data sample

The dataset [1] contains over 16 600 dates when the number of sunspots were recorded. Two columns - date and sunspot count, were used for displaying the solar cycles. To polish the data set, missing rows were excluded and for clear depiction of the peaks, the 13-month centered average was used. This technique is used to target a specific month withing a specific year, it smoothens out the data so the true peak is evidently depicted. After the centered average was implemented it was easier to locate the solar maximums, marked as solar peaks. This can be seen in the first graph in the section "Results and Discussion".

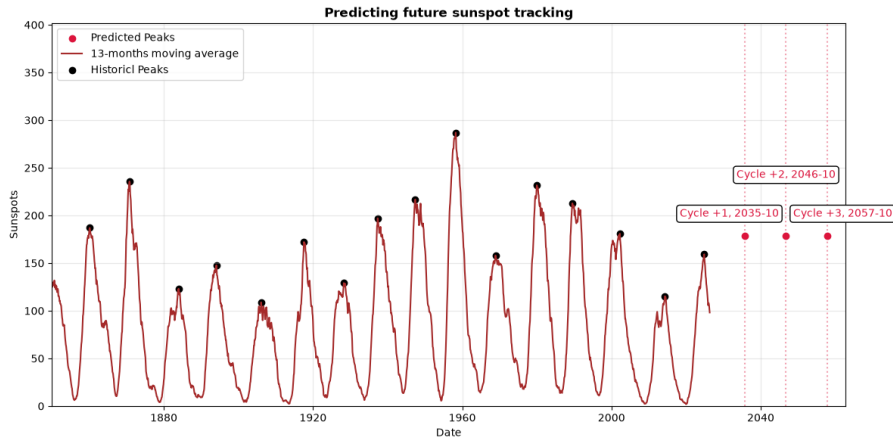
For the second part of the research, another 3 future cycles were to be predicted. Even though it is a known fact that solar cycles repeat every eleven years, the purpose was to see if the theoretical values match with the predictions done through python. The approach was to take an average of a cycle, convert the value from years to days and add this converted value to the date of the last detected peak, represented in the second graph in the section "Results and Discussion".

Results and discussion

After analyzing over 16 600 lines of data, overall 16 peaks were detected, which is in accordance with the mathematical predictions. From 1850 to 2026, 176 years pass, divided by 11 years as the standard length of a cycle, gives exactly 16 peaks for exposure. Precisely, the average length of a cycle was 10.97 years with a standard deviation of 1.1 years. In the process, an average amplitude of a cycle was calculated. The unit is in sunspots, as more sunspots are detected the more likely is the cycle in solar maximum. With this considered, an average amplitude reaches the value of 178.89 sunspots with the standard deviation of 48.11 sunspots.



After this, the predictions can be programmed. The methodology and the approach were discussed in the section "Data Sample". It produced three dates - 2035, 2046 and 2057. To highlight it, the resulting dates are marked with red dots and dashed line. This corresponds with the initial predictions and hypothesis. Unlimited number of future cycles can be predicted through the mentioned way, however for this project just three were chosen for a clear demonstration.



This research project accomplished the desired outcome, and to readdress the research question, the next solar cycle can be predicted through python in VS code with plots representing the dates. As expansion, it would be beneficial to look at the solar cycles through number of sunspots not only through years between solar maximums and predict how many sunspots can the future cycles potentially have.

References

- [1] World Data Center SILSO. *Daily Sunspots dataset*. Available at: <https://www.kaggle.com/datasets/patrickfleith/daily-sunspots-dataset>.
- [2] What is the Solar Cycle? *NASA Science*. Available at: <https://spaceplace.nasa.gov/solar-cycles/en/>
- [3] How often does the sun's magnetic field flip? *Science Questions with Surprising Answers*. Available at: <https://www.wtamu.edu/~cbaird/sq/2013/08/07/how-often-does-the-suns-magnetic-field-flip/>